Sequences and Series

ANSWER KEY

Section 1: Limits of Sequences

1 Find the limit of the sequence $a_n = \frac{3n^2 - 5n}{2n^2 + 7}$ as $n \to \infty$.

a. $3/2$ ✓

b. 0

c. ∞

d. $3/7$

Divide numerator and denominator by n^2: $\lim_{n\to\infty} \frac{3 - 5/n}{2 + 7/n^2} = \frac{3}{2}$.

2 Determine if the sequence $a_n = \frac{(-1)^n}{n}$ converges, and if so, find its limit.

a. **Converges to** 0 ✓

b. Diverges (oscillates)

c. Converges to 1

d. Converges to -1

Since $|a_n| = 1/n \to 0$ as $n \to \infty$, the sequence converges to 0 by the Absolute Value Theorem for sequences.

3 Find the limit of the sequence $a_n = \left(1 + \frac{2}{n}\right)^n$ as $n \to \infty$.

a. e^2 ✓

b. e

c. 1

d. ∞

Using the standard exponential limit $\lim_{n\to\infty} (1 + x/n)^n = e^x$, with $x = 2$ we get e^2.

Section 2: Geometric Series

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4 Find the sum of the geometric series $\sum_{n=1}^{\infty} 3\left(\frac{2}{5}\right)^{n-1}$.

a. 5 ✓

b. 15/2

c. 3

d. 15

First term $a = 3$, common ratio $r = 2/5$. Since $|r| < 1$, sum $S = \frac{a}{1-r} = \frac{3}{1-2/5} = \frac{3}{3/5} = 5$.

5 Determine the convergence of $\sum_{n=0}^{\infty} 4\left(-\frac{3}{2}\right)^{n}$.

a. **Diverges because** $|r| = 3/2 \geq 1$ ✓

b. Converges to $-8/5$

c. Converges to $8/5$

d. Converges to 0

The common ratio is $r = -3/2$. Since $|r| = 1.5 \geq 1$, the geometric series diverges.

6 Express the repeating decimal $0.\overline{27} = 0.272727\ldots$ as a simplified fraction using a geometric series.

a. 3/11 ✓

b. 27/100

c. 9/33

d. 27/99 is 3/11

$a = 27/100, r = 1/100$. $S = \frac{27/100}{1-1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}$.

Section 3: Telescoping Series

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7 Find the sum of the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1 ✓

b. 1/2

c. 2

d. Diverges

Partial fractions: $\frac{1}{n} - \frac{1}{n+1}$. Partial sum
$S_k = (1 - 1/2) + (1/2 - 1/3) + \ \dots\ + (1/k - 1/(k+1)) = 1 - \frac{1}{k+1} \to 1$ as $k \to \infty$.

8 Find the sum of $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$.

a. 1 ✓

b. 0

c. ∞

d. $\sqrt{2}$

$S_k = (1 - 1/\sqrt{2}) + (1/\sqrt{2} - 1/\sqrt{3}) + \ \dots\ + (1/\sqrt{k} - 1/\sqrt{k+1}) = 1 - 1/\sqrt{k+1} \to 1.$

9 Evaluate $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$.

a. **Diverges to** ∞ ✓

b. 0

c. 1

d. $\ln 2$

$S_k = \sum_{n=1}^{k} [\ln(n+1) - \ln(n)] = \ln(k+1) - \ln(1) = \ln(k+1) \to \infty$ as $k \to \infty$.

Section 4: Divergence Test (nth-Term Test)

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10 What does the nth-Term Test for Divergence state?

a. **If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges ✓**

b. If $\lim_{n\to\infty} a_n = 0$, then $\sum a_n$ converges

c. If $\sum a_n$ diverges, then $\lim_{n\to\infty} a_n \neq 0$

d. If $\lim_{n\to\infty} a_n = 0$, the test proves divergence

The test states: if the terms do not approach 0, the series MUST diverge. If $\lim a_n = 0$, the test is inconclusive.

11 Apply the Divergence Test to $\sum_{n=1}^{\infty} \frac{n}{2n+1}$.

a. **Diverges because $\lim_{n\to\infty} a_n = 1/2 \neq 0$ ✓**

b. Converges to $1/2$

c. Inconclusive

d. Converges by p-series

$\lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} \neq 0$. Since the limit of the terms is non-zero, the series diverges.

12 Why is the Divergence Test inconclusive for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. **Because $\lim_{n\to\infty} \frac{1}{n} = 0$, which does not guarantee convergence ✓**

b. Because $1/n$ is negative

c. Because the terms oscillate

d. Because it is a geometric series

$\lim_{n\to\infty} 1/n = 0$. Having terms approach 0 is a necessary condition for convergence, but not a sufficient one.

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